Milestone 3 – Model Training and Evaluation

1. Introduction

Milestone 3 focuses on training and evaluating a deep learning model for document scanner source identification. In this milestone, a Convolutional Neural Network (CNN) is employed to automatically learn scanner-specific visual patterns from scanned document images.

The goal is to identify the scanner model used to digitize a document, based on inherent visual artifacts such as texture, noise patterns, and resolution characteristics introduced during the scanning process.

2. **Objective**

The objectives of this milestone are:

To train a CNN-based model for document scanner source identification

To classify scanned document images into different scanner categories

To evaluate the model's performance using standard deep learning metrics

3. **Dataset Description**

The dataset used in Milestone 3 consists of scanned document images produced using different scanner models.

Each image is labeled according to the scanner used to generate it.

3.2 Dataset Characteristics

Multi-class classification problem
Images scanned at 150 DPI and 300 DPI
Images resized to 224 × 224 pixels

Dataset split:

80% Training
20% Validation

4. **Preprocessing and Data Augmentation**

4.1 Preprocessing

To ensure consistency across inputs, the following preprocessing steps were applied:
Pixel value normalization using rescaling (1/255)
Resizing all images to a fixed resolution compatible with the CNN model

4.2 Data Augmentation

Image rotation
Zoom transformations
Horizontal flipping

The validation dataset was only rescaled and not augmented, ensuring unbiased evaluation.

5. Model Architecture

A Convolutional Neural Network (CNN) was designed to extract scanner-specific features from document images.

5.1 Architecture Overview

The model consists of:
Convolutional layers with ReLU activation to learn local visual patterns
MaxPooling layers for spatial downsampling
A Flatten layer to convert feature maps into a feature vector
Fully connected Dense layers for classification
Dropout layers to reduce overfitting
An output layer that predicts the scanner category of the input document
The architecture is designed to balance performance and computational efficiency.

6. Training Configuration

The CNN model was trained with the following configuration:
Optimizer: Adam
Loss Function: Categorical Crossentropy
Epochs: 10

Training progress was monitored using training and validation accuracy and loss metrics.

7. Model Evaluation

The trained model was evaluated using the validation dataset.

7.2 Results

The model showed consistent improvement across training epochs
Validation accuracy reached approximately 95%

8. **Performance Summary**

Task: Document scanner source identification
Model: Convolutional Neural Network (CNN)
Validation Accuracy: ~95%
Generalization: Stable across scanner categories

The results indicate that the CNN successfully learned scanner-dependent visual characteristics from document images.